

Data Collection and Analysis: PGR215

Data Engineering

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Topics for Today

What is Data Engineering?

Data Professionals

What is a Data Pipeline?

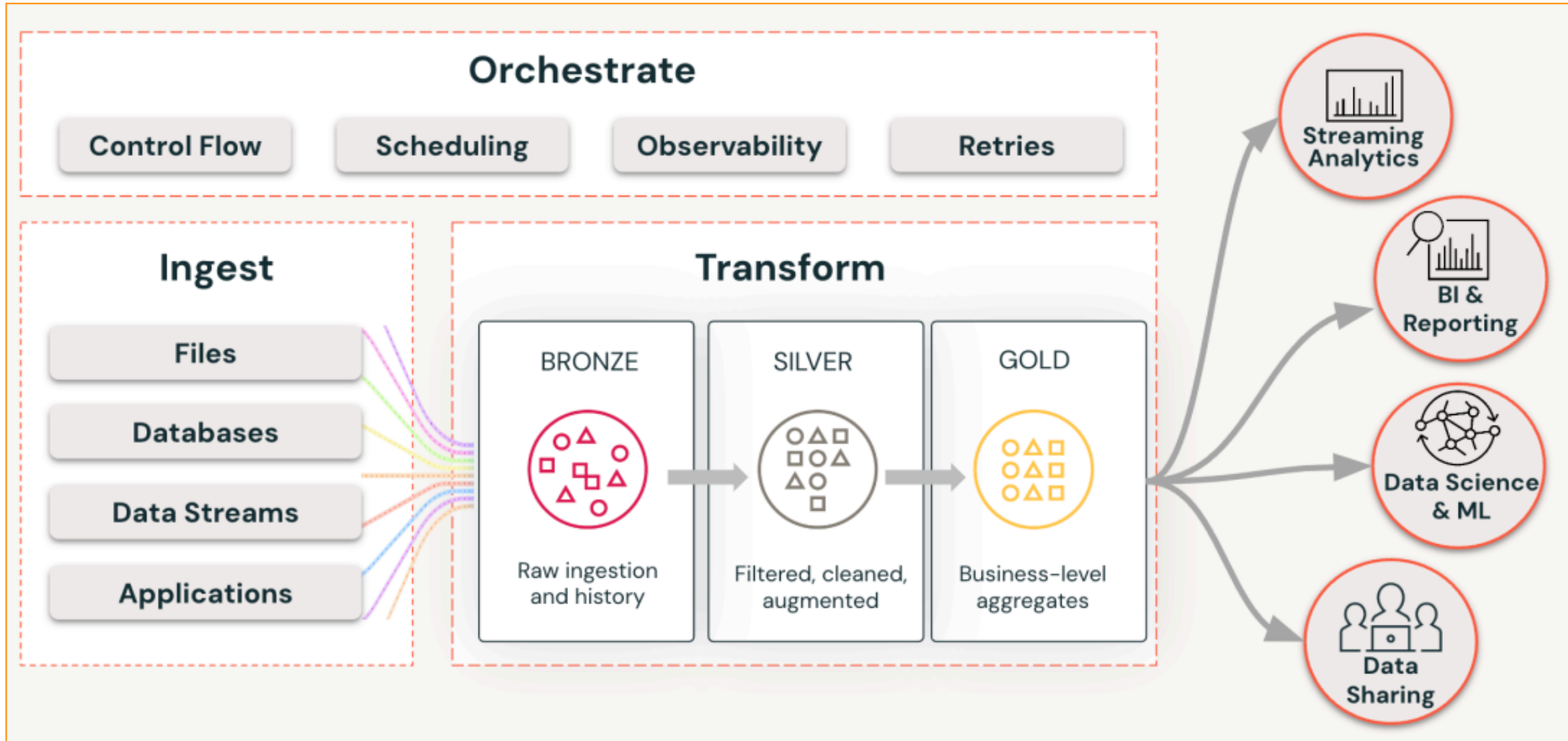
Different Data Files and Sources

ETL vs. ELT

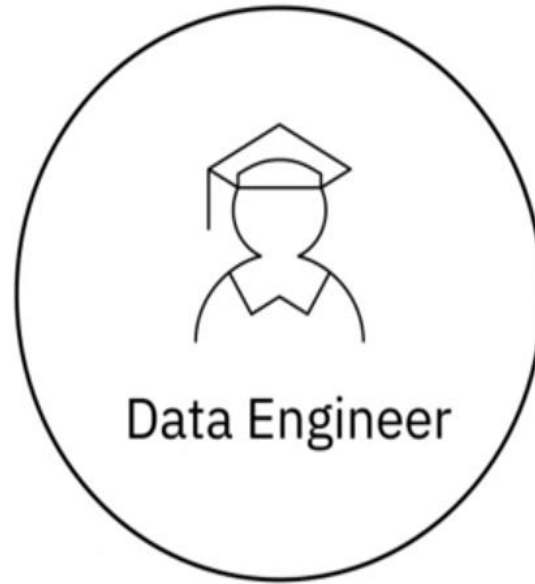
What is Data Engineering?

What is Data Engineering?

Data engineering is the practice of taking raw data from a data source and processing it such that it's stored and organized for a downstream use case such as data analytics, business intelligence (BI) or machine learning (ML) model training. In other words, it's the process of preparing data so value can be extracted from it.



Data Professionals



Data Professional excited about engineering



Statistician



Data
Scientist



Data
Analyst



BI
Analyst

Working professional with a technical role



IT
Support



Software
Tester



Programmer

Data Engineer



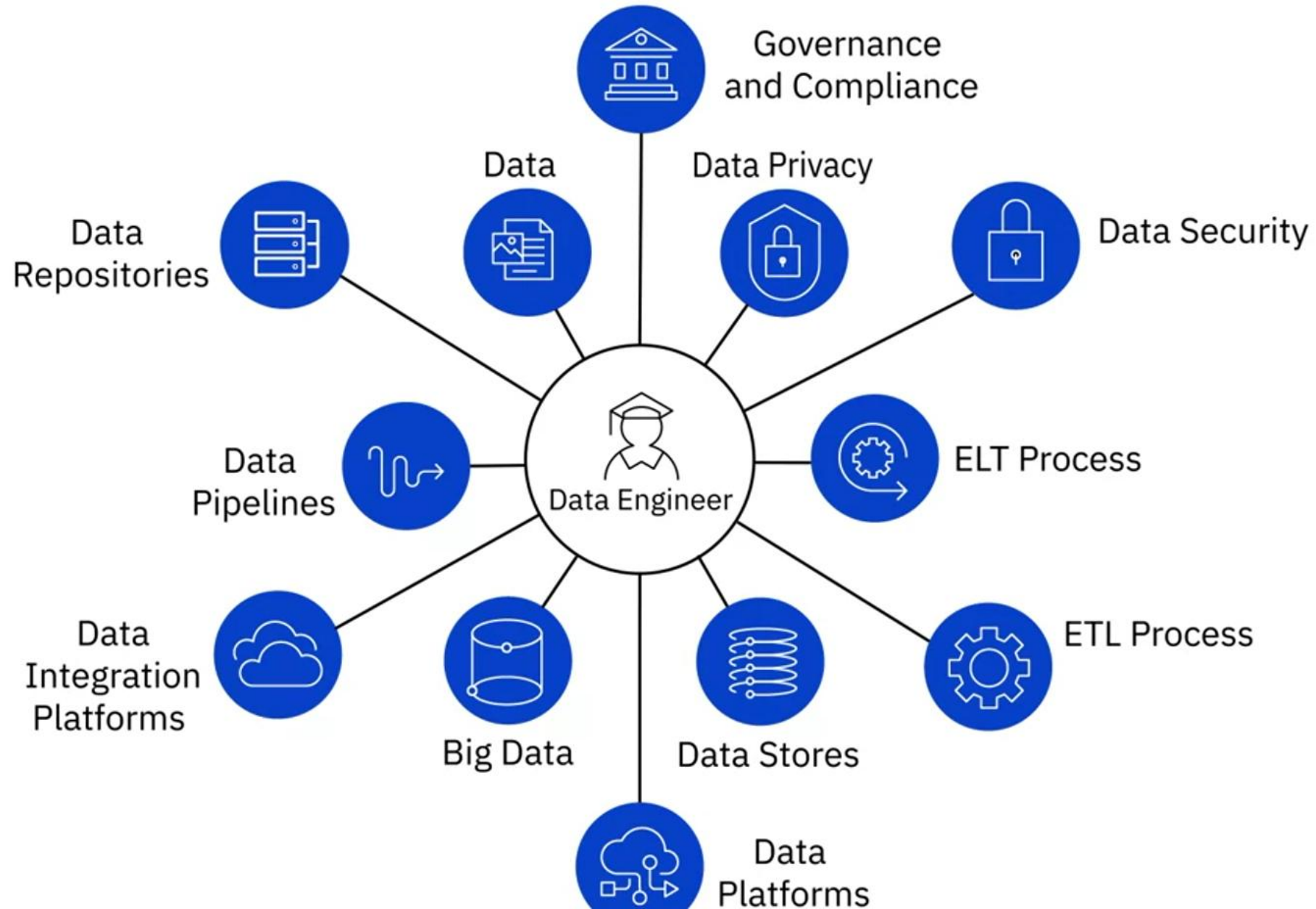
Data Engineers work within the data ecosystem to:

- Extract, integrate, and organize data from disparate sources
- Clean, transform, and prepare data

Skills:

- Good knowledge of programming
- Sound knowledge of systems and technology architectures
- In-depth understanding of relational databases and non-relational data stores

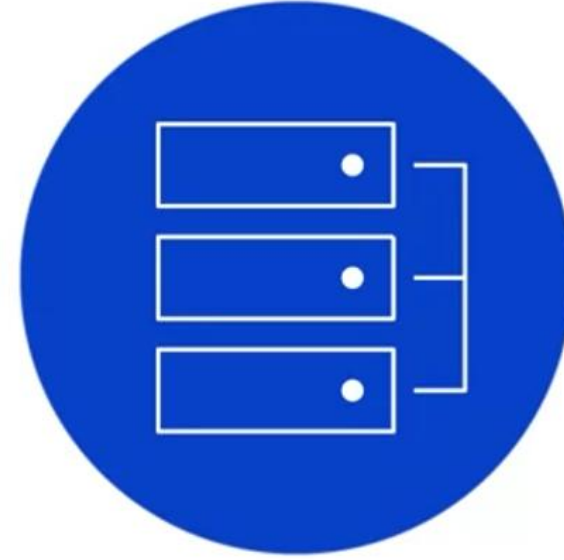
Data Engineer Skills



The value we derive from data depends upon:



Accuracy of data



Accessibility of data
when we need it

Role of Data Engineer

Data Analyst



Responsibilities of a Data Analyst:

- Inspect and clean data for deriving insights
- Identify correlations, find patterns, and apply statistical methods to analyze and mine data
- Visualize data to interpret and present the findings of data analysis



“Are the users’ search experiences generally good or bad with the search functionality on our site?”

“What is the popular perception of people regarding our rebranding initiatives?”

“Is there a co-relation between sales of one product and another?”

Skills:

- Good knowledge of spreadsheets, writing queries, and using statistical tools to create charts and dashboards
- Programming skills
- Strong analytical and story-telling skills

Data Scientist



Responsibilities of a Data Scientist:

- Analyze data for actionable insights
- Create predictive models using Machine Learning and Deep Learning



“How many new social media followers am I likely to get next month?”

“What percentage of my customers am I likely to lose to competition in the next quarter?”

“Is this financial transaction unusual for this customer?”

Skills:

- Knowledge of Mathematics and Statistics
- Understanding of programming languages, databases, and building data models

Business Analyst and BI Analyst



Business Analysts leverage the work of Data Analysts and Data Scientists to look at possible implications for their business and the actions they need to take or recommend.



Data Analysts



Data Scientists



BI Analysts

- Focus on market forces and external influences that shape their business
- Organize and monitor data on different business functions
- Explore data to extract insights and actionables that improve business performance

Summary

- Data Engineering convert raw data into usable data
- Data Analytics use this data to generate insights
- Data Scientists use Data Analytics and Data Engineering to predict the future using data from the past
- Business Analysts and Business Intelligence Analysts use these insights and predictions to drive decisions that benefit and grow their business

How Data Teams Work Together

E-commerce company improving customer retention

Data Engineer



Builds a data pipeline to collect, clean, and store customer data

Output:
Organized data

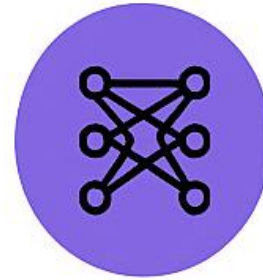
Data Analyst



Analyzes purchasing patterns and customer churn rates

Output:
Insights

Data Scientist



Builds a model to predict the likelihood of customer churn

Output:
Predictions

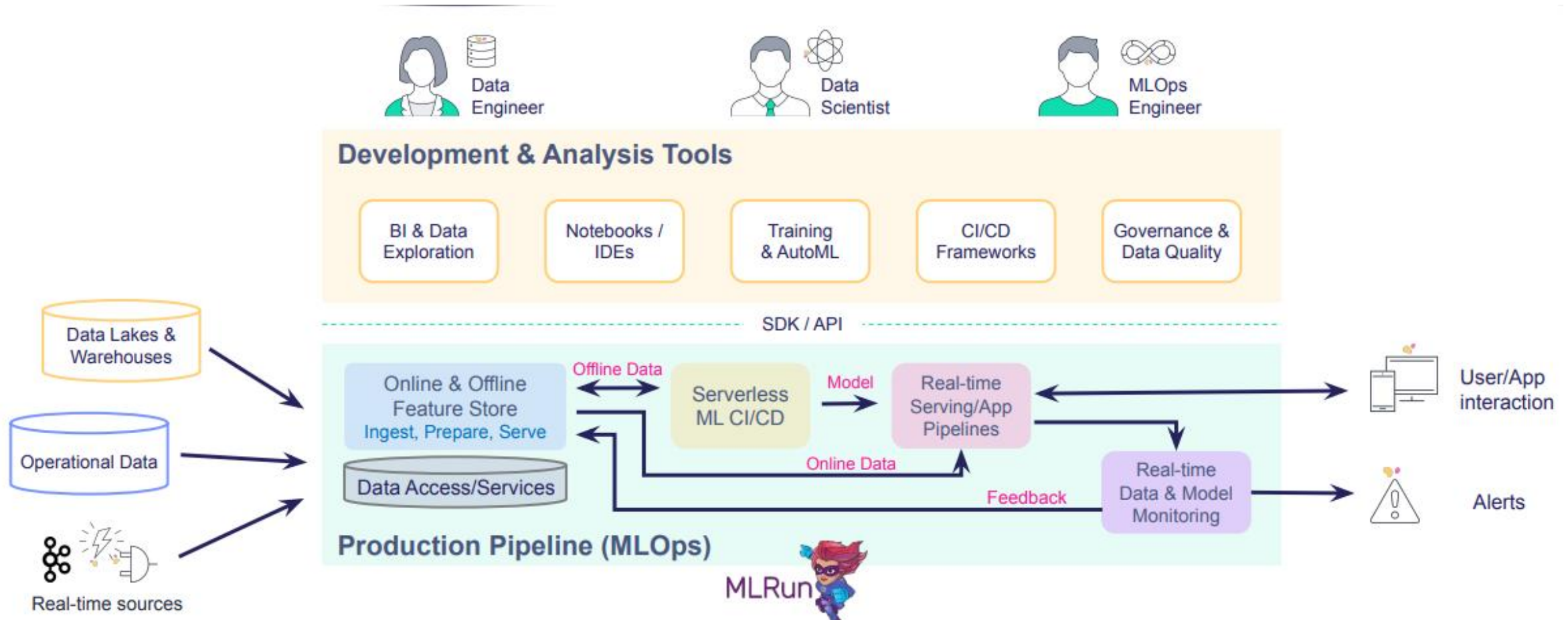
Business Analyst



Develops strategies to improve customer retention

Output:
Strategic decisions

Data Engineer vs. Data Scientist vs. ML Engineer



Data Engineer vs. Data Scientist vs. ML Engineer

Aspect	Data Engineer	Data Scientist	ML Engineer
Key Responsibilities	- Design and maintain data pipelines and infrastructure - Develop ETL processes for data ingestion and transformation - Ensure data quality, storage, and accessibility	- Analyze data to extract insights and inform decision-making - Develop statistical models and algorithms - Perform data visualization and reporting	- Implement and deploy machine learning models in production - Optimize models for scalability and performance - Collaborate with data scientists and engineers for model integration
Required Skills	- SQL, Python, Java/Scala - Big Data technologies (e.g., Hadoop, Spark) - Database management (e.g., SQL, NoSQL)	- Statistics, Data Analysis - Python/R, SQL - Machine learning & deep learning frameworks (e.g., Scikit-Learn, TensorFlow)	- Python, Machine learning - Model deployment frameworks (e.g., Flask, Docker) - Deep learning frameworks (e.g., TensorFlow, PyTorch)
Typical Tools	- Apache Spark, Hadoop, Kafka - AWS/GCP/Azure - ETL tools (e.g., Airflow)	- Jupyter, Pandas, Matplotlib - Scikit-Learn, TensorFlow - Data visualization tools (e.g., Tableau)	- TensorFlow, PyTorch, MLflow - Docker, Kubernetes - Cloud ML tools (AWS SageMaker, Google AI Platform)
Focus Area	Data storage, processing, and pipeline management	Data analysis, model building, and business insights	Model deployment, optimization, and scalability

Trivia Time

1. What is the primary role of a data engineer in the context of data processing and analysis?

a) Predict future trends

★ b) Convert raw data into usable data

c) Analyse data for insights

d) Use data to make business decisions

2. How do data analytics professionals utilize the data prepared by data engineers?

a) Use data to make business decisions

b) Predict future outcomes

★ c) Generate insights

d) Create machine learning models

3. What is the role of data scientists in the data workflow, and how do they leverage the work of data engineers and data analysts?

a) Convert raw data into usable data

★ b) Use data analytics and engineering to predict the future using past data

c) Visualize insights for stakeholders

d) Make decisions based on generated insights

Trivia Time

4. How do business analysts and business intelligence analysts utilize the insights and predictions provided by data professionals?

a) Generate more data

★ b) Make decisions that benefit and grow the business

c) Create predictive models

d) Manage communication networks

5. Which role in the data ecosystem is responsible for ensuring that data is in a structured and usable form for downstream processes?

a) Data scientists

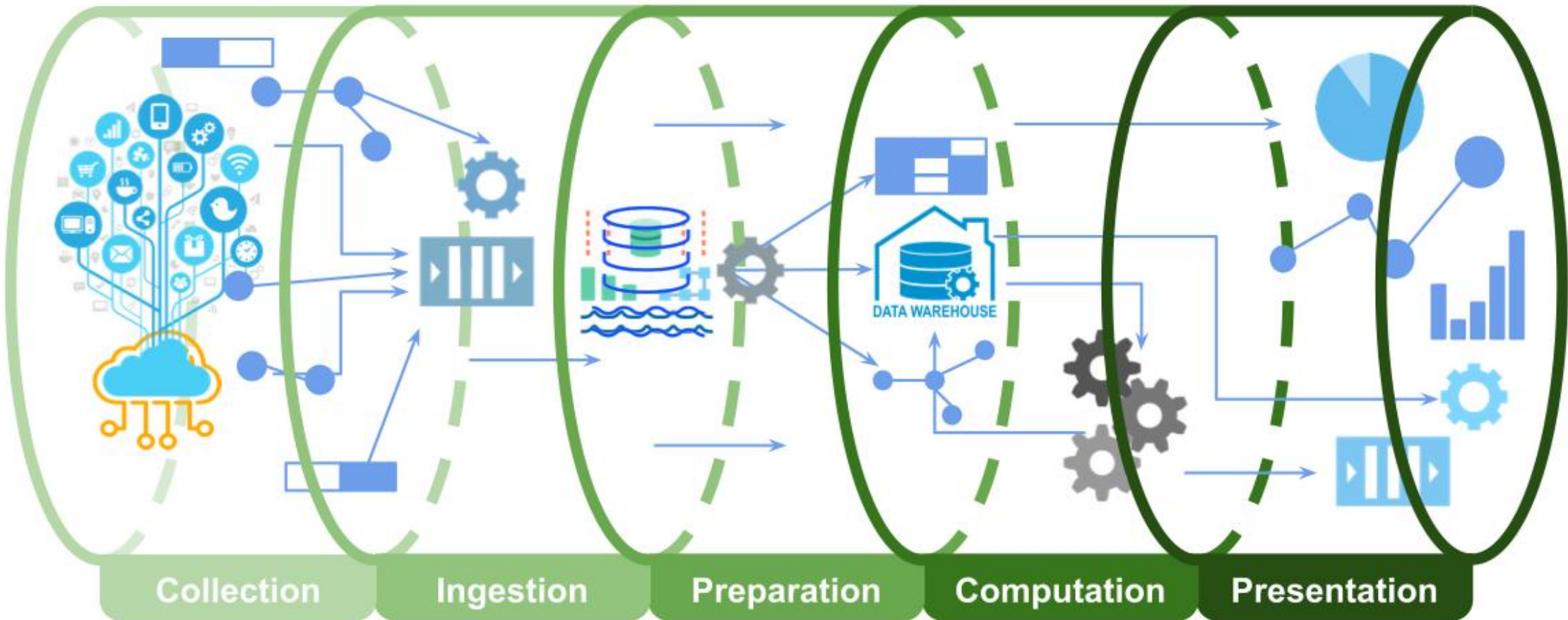
★ b) Data engineers

c) Business analysts

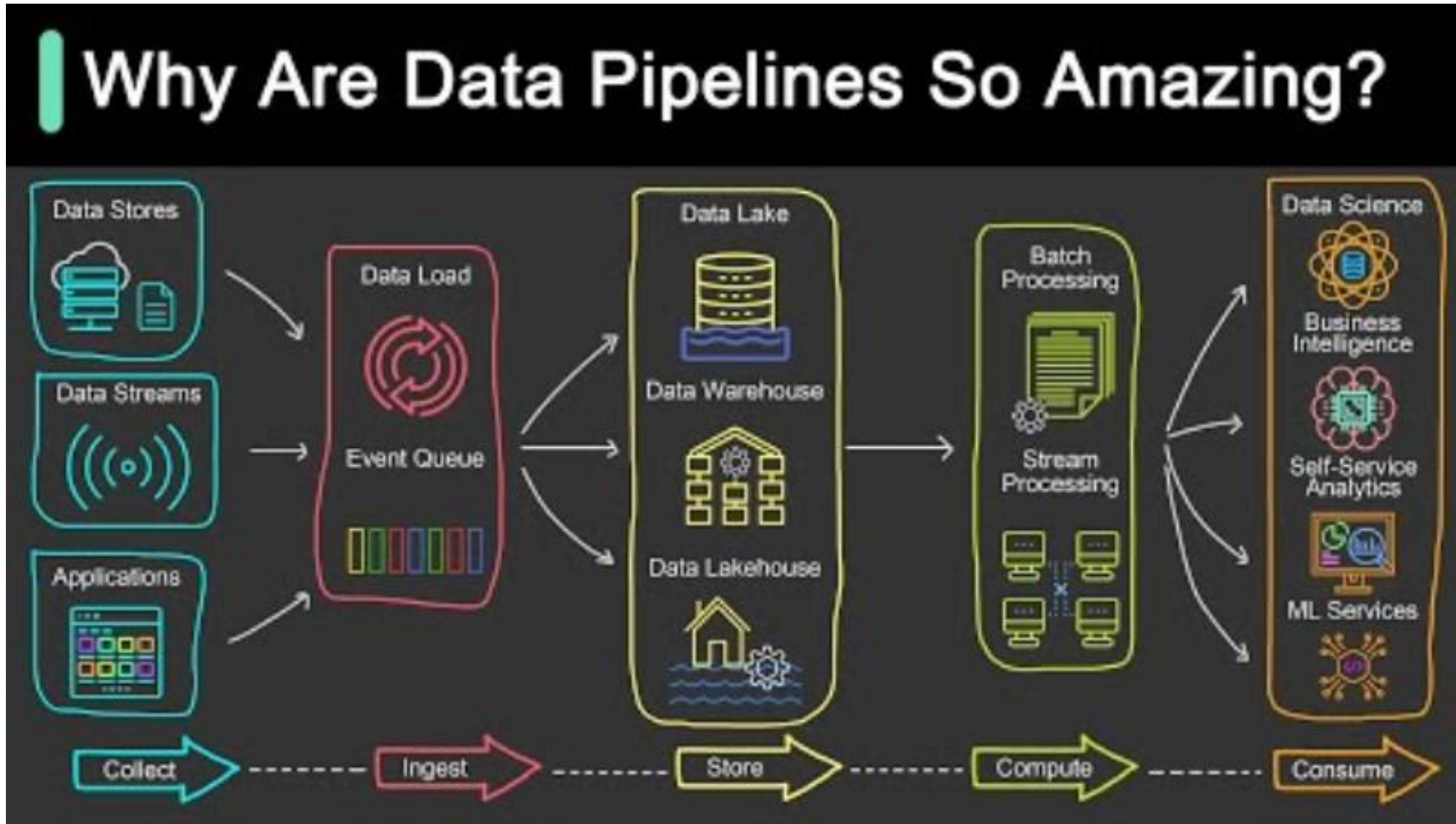
d) Data visualization experts

What is a Data Pipeline?

Pipeline is a series of connected processes, such that output of one process is the input of the next.



What is a Data Pipeline?



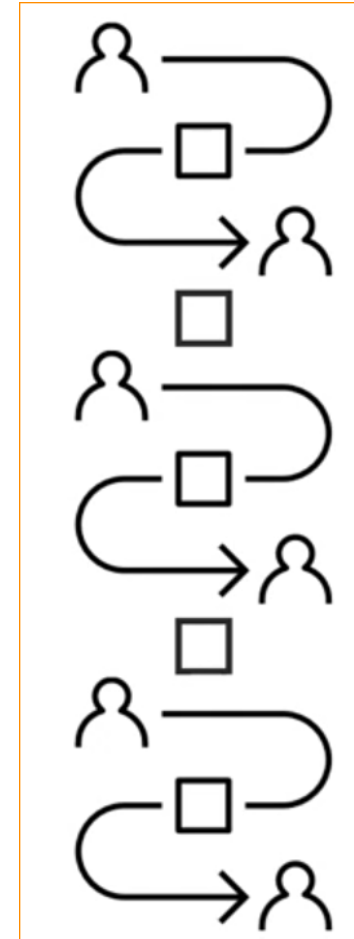
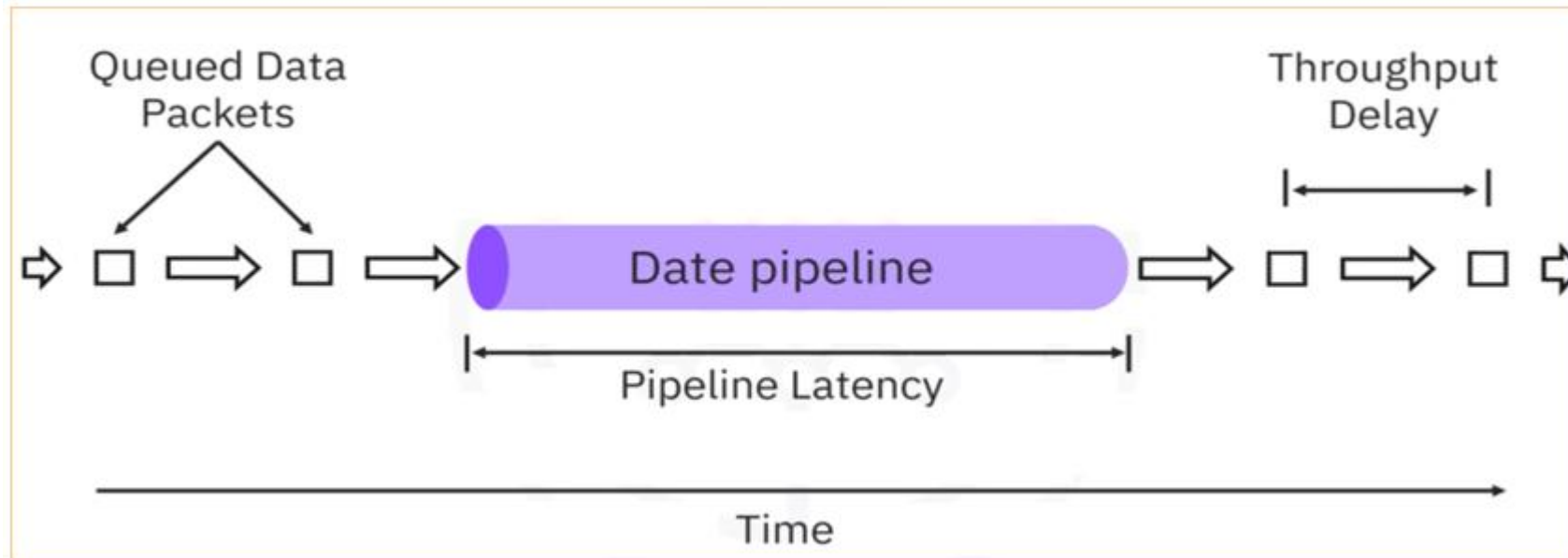
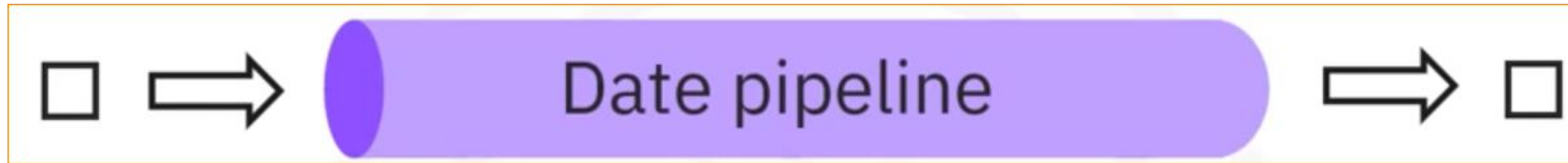


BREAK
15MINS

What is a Data Pipeline?

Purpose: Move data from one place or form to another.

Any system that extracts, transforms and loads data.



Stages of Data Pipeline Process



Data extraction

Collect raw data from various sources (databases, APIs, files) as the first step in the pipeline.



Loading into
destination
facility

Load the transformed data into a target storage system (e.g., data warehouse, data lake) for access and analysis.



Data ingestion

Transfer extracted data into the pipeline, either in real-time or as batch data, for further processing



Scheduling or
triggering

Set schedules or triggers to automate and control the execution of data pipeline processes.



Transformation
stages

Apply transformations (cleaning, filtering, enrichment) to prepare and standardize data for analysis.



Monitoring

Track pipeline performance and data quality to ensure data flows smoothly without errors.



Maintenance
and
optimization

Continuously maintain and optimize the pipeline for improved efficiency, scalability, and reliability.

Pipeline Monitoring Considerations



Latency



Throughput



Warnings,
errors, failures



Utilization rate



Logging and alerting
system

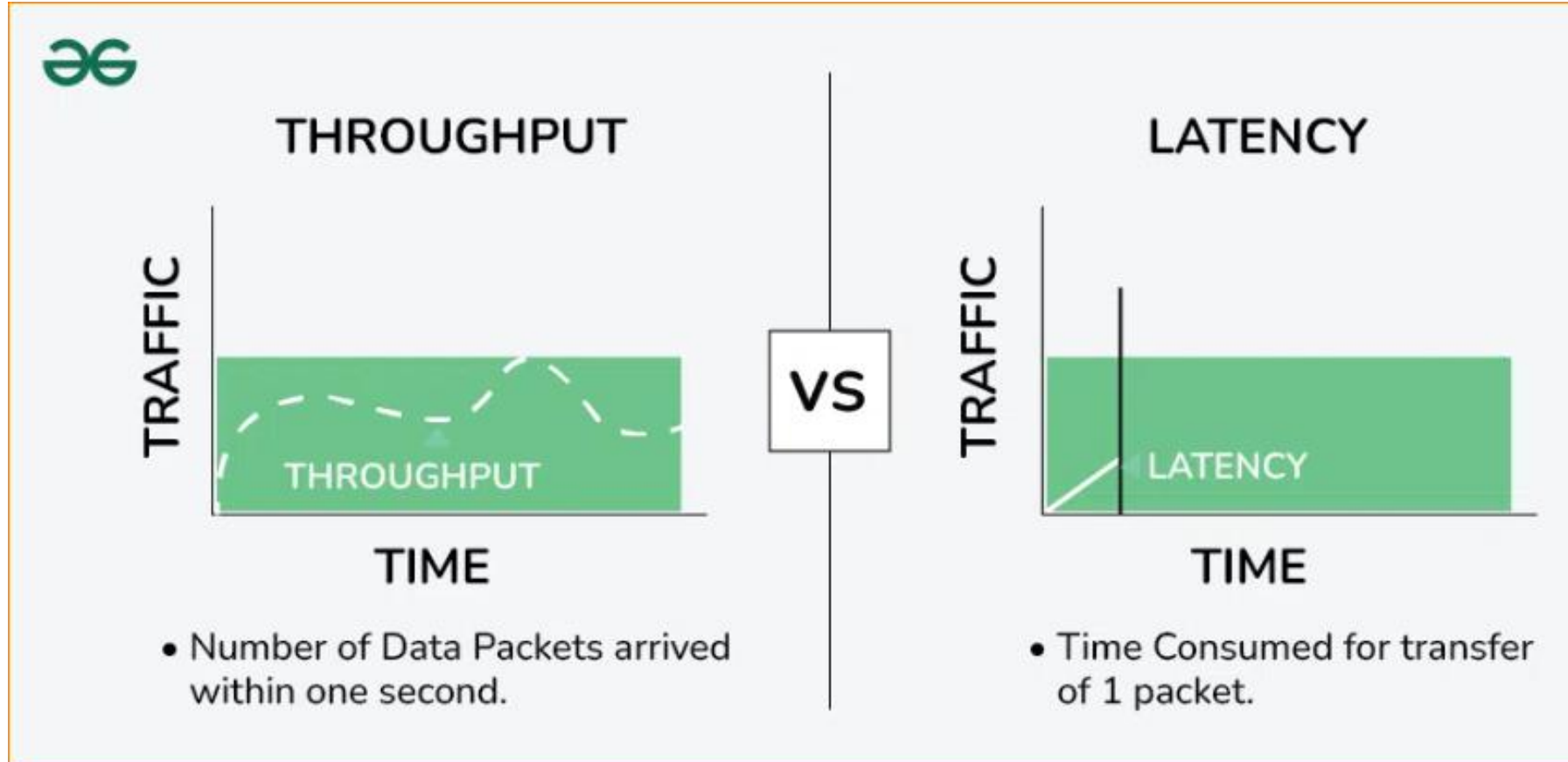
Data Pipeline Performance

Performance Measure: Throughput

Throughput is the **volume of data** that the pipeline can process in a given period, usually measured in records per second or bytes per second. Higher throughput is essential for handling large data volumes efficiently.

Performance Measure: Latency

Latency refers to the **time delay** it takes for data to travel through the pipeline from source to destination. Lower latency is critical for real-time applications that require immediate data insights.



Batch vs. Stream Data Pipelines

Batch Data Pipeline

-  Operate on batches of data
-  Usually run periodically – hours, days, weeks apart
-  Can be initiated based on data size or other triggers
-  When the latest data isn't needed
-  Typical choice when accuracy is critical

Stream Data Pipeline

-  Ingest data packets in rapid succession
-  For real-time results
-  Records/events processed as they happen
-  Event streams can be loaded into storage
-  Users publish/subscribe to event streams

Use Cases of Batch Data Pipeline



Data backups



Transaction
history loading



Billing and order
processing



Data modeling



Forecasting sales
or weather



Retrospective data
analysis



Diagnostic
medical image
processing

Use Cases of Streaming Data Pipeline



Watching movies, listening to music or podcasts



Social media feeds, sentiment analysis



Fraud detection



User behavior, targeted advertising



Stock market trading



Real-time product pricing



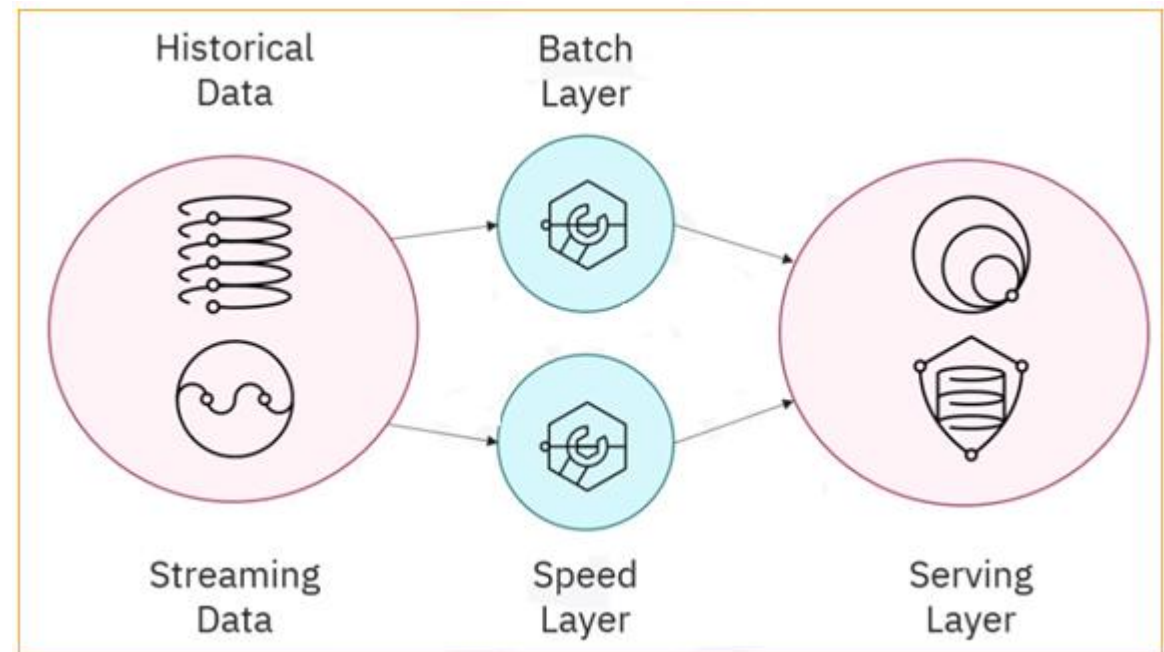
Recommender systems

Batch vs. Stream Requirements

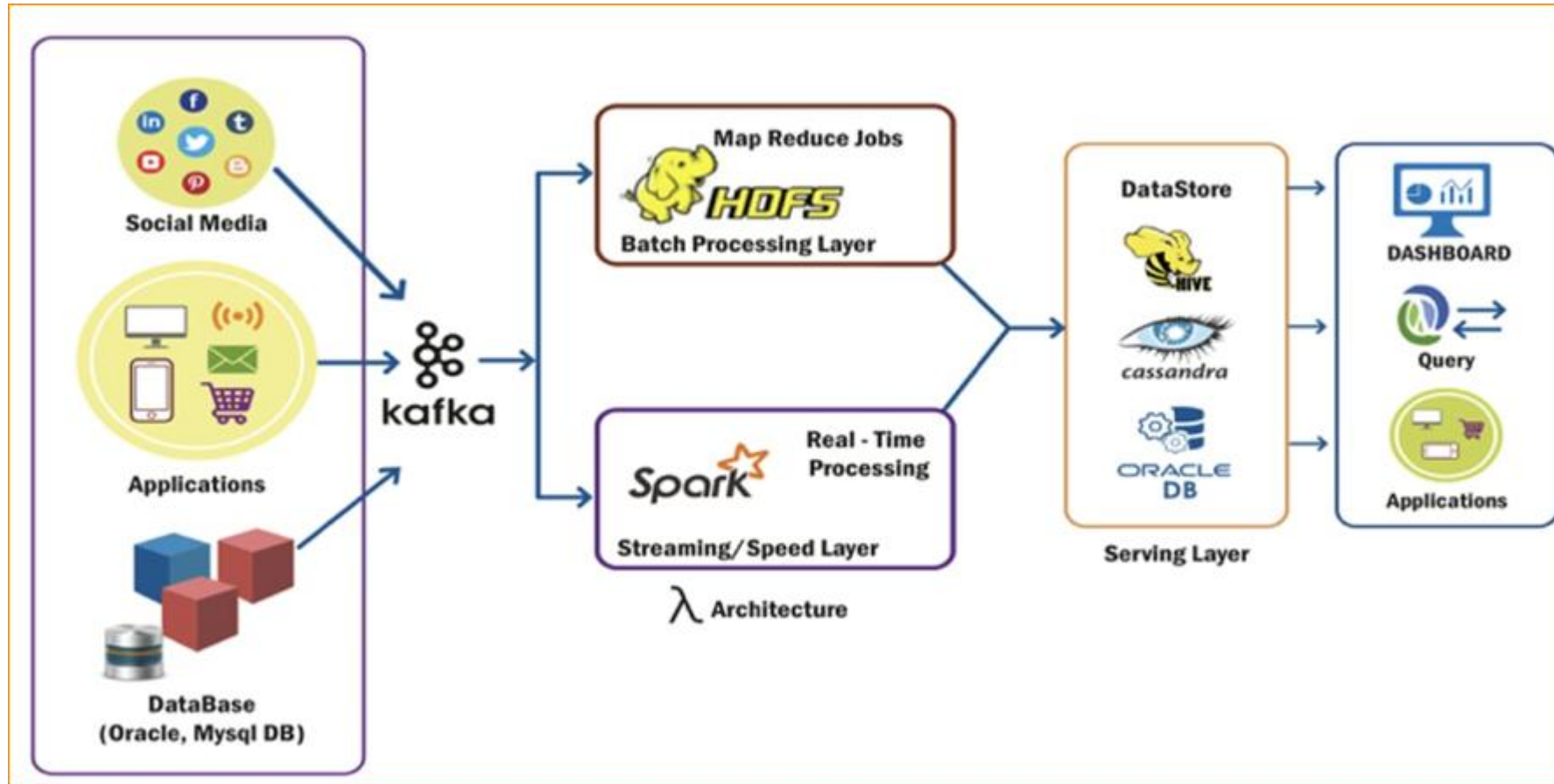


Lambda Architecture

Lambda architecture is a data-processing architecture designed to handle massive quantities of data by using both **batch processing** and **real-time processing** methods. It was designed to provide a scalable and fault-tolerant system for managing big data



Lambda Architecture



Batch Layer: Deals with historical data and can handle massive datasets, ensuring durability and accuracy over time.

Speed Layer: Complements the batch layer by providing real-time results on the most recent data, ensuring that the system can respond quickly to changes.

Serving Layer: Aggregates the outputs from both the batch and speed layers to allow users to query both historical and real-time data.

Lambda Architecture

Layer	Description	Key Technologies	Output
Batch Layer	- Processes large volumes of data in batch mode. - Stores raw data immutably. - Provides fault tolerance and complete view of historical data.	- Hadoop - Apache Spark	Batch Views: Pre-computed views of large-scale historical data.
Speed Layer	- Handles real-time data processing. - Provides low-latency insights. - Complements batch layer by processing recent, unprocessed data.	- Apache Kafka - Apache Storm - Apache Flink	Real-time Views: Instantaneous results based on recent data.
Serving Layer	- Combines results from both the batch and speed layers. - Provides a unified view for querying.	- NoSQL databases (e.g., Cassandra, HBase) - Elasticsearch	Unified Results: A combination of historical (batch) and real-time views for querying.

Trivia Time

What is the main purpose of a data pipeline?

- ☐ Storage
- ☐ Loading
- ☒ Move data from one place or form to another
- ☐ Data container

Which of the following is the main reason to use Lambda architecture instead of Micro-batch, Streaming, or Batch data pipelines?

- ☒ Access to earlier data and speed is important
- ☐ Accuracy is critical
- ☐ Help with load balancing
- ☐ Records are processed immediately

Which of the following data pipeline processes keep the pipeline running smoothly?

- ☐ Ingestion
- ☒ Maintenance and Optimization
- ☐ Monitoring
- ☐ Scheduling and Triggering

Trivia Time

Which of the following data pipelines corresponds with the fraud detection use case?

- ☐ Batch data pipeline
- ☐ Micro-batch data pipeline
- ☒ Streaming data pipeline
- ☐ Lambda architectures

Which of the following data pipeline use cases is the simplest?

- ☐ Transactional record movement
- ☒ File backup
- ☐ Raw data preparation
- ☐ Send/receive messages

Which of the following common features of modern ETL and ELT products is known as "no-code"?

- ☐ Security
- ☐ Data crawling
- ☐ Fully automated
- ☒ Drag-and-drop

Trivia Time

How does data flow through pipelines?

- ☒ Data packets
- ☐ Files
- ☐ Software processes
- ☐ Processing threads

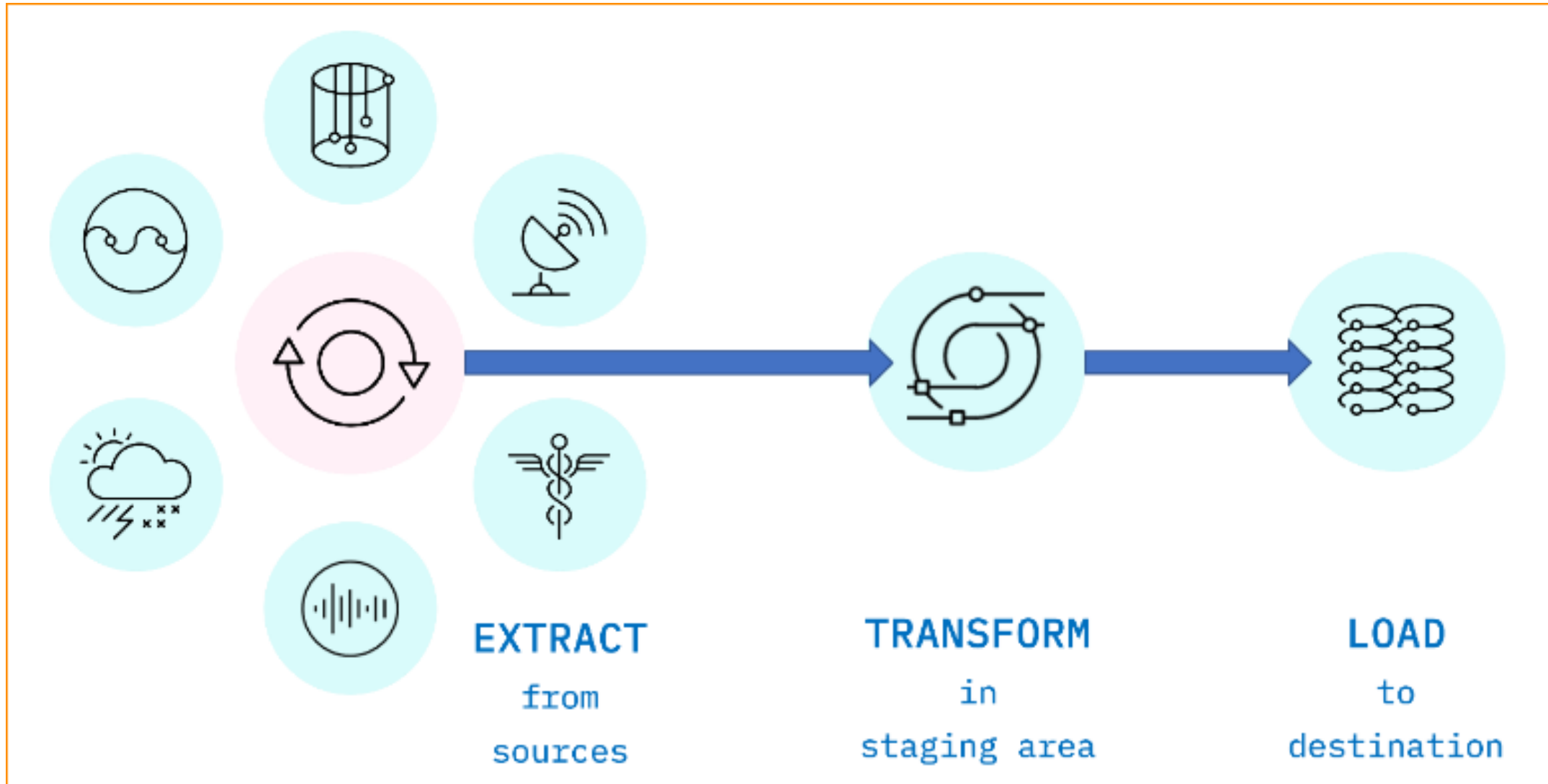
Which of the following pipeline monitoring considerations affects the amount of data that passes through the pipeline over time?

- ☒ Throughput
- ☐ Latency
- ☐ Logging and alerting system
- ☐ Utilization

Latency is the total time it takes for a single packet of data to pass through the pipeline. Which of the following limits latency?

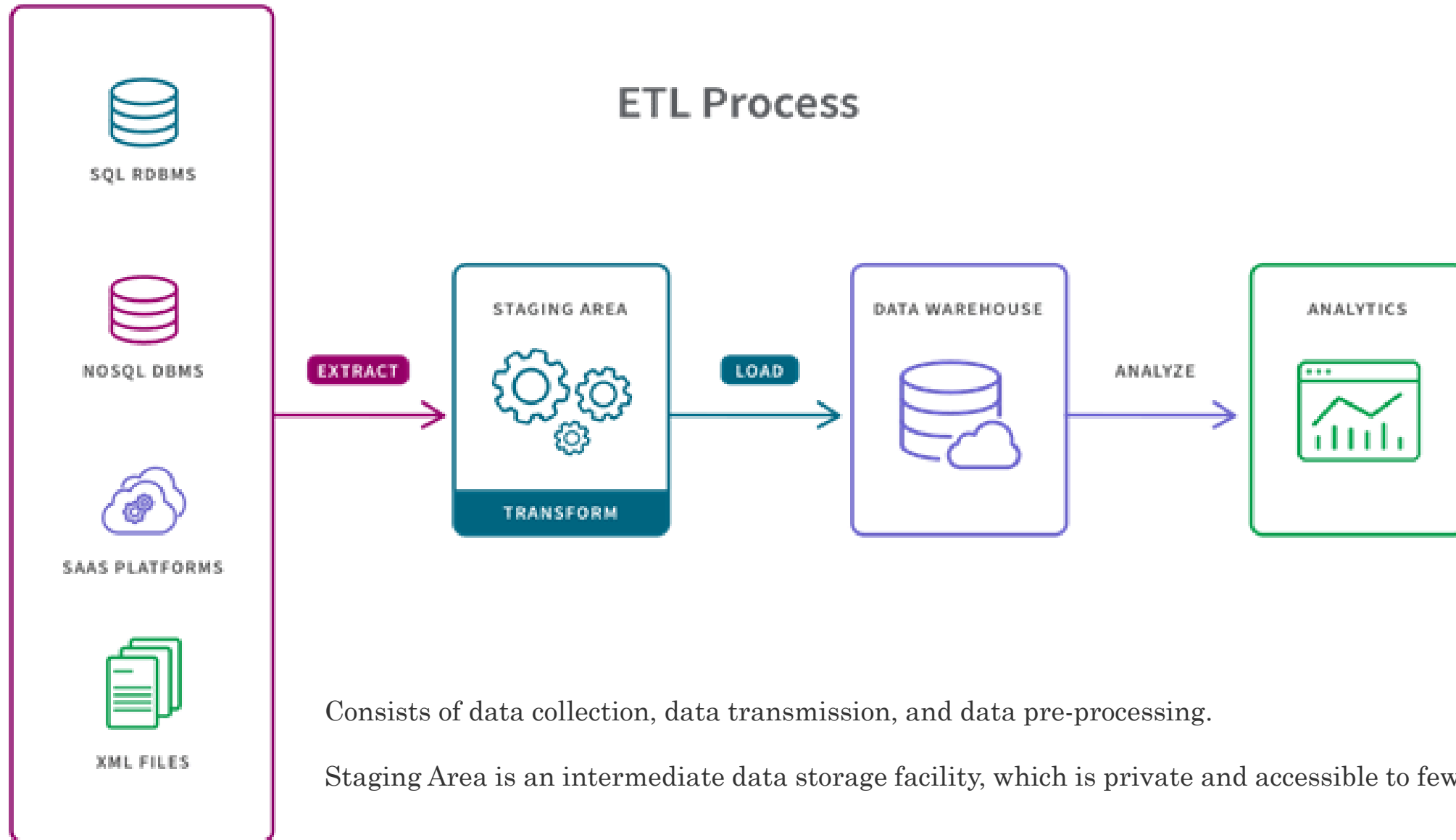
- ☐ Small data packets
- ☒ Slowest process
- ☐ Bad data
- ☐ Data leak

ETL Data Pipeline



ETL stands for Extract, Transform, and Load, and refers to the process of curating data from multiple sources, conforming it to a unified data format or structure, and loading the transformed data into its new environment.

ETL Data Pipeline



Data Staging

- Between transformations, data is often staged—written in a temporary state to a suitable location, often either cloud storage or an intermediate table. Staged data may be kept temporarily or indefinitely, but it plays a crucial role in transformation.

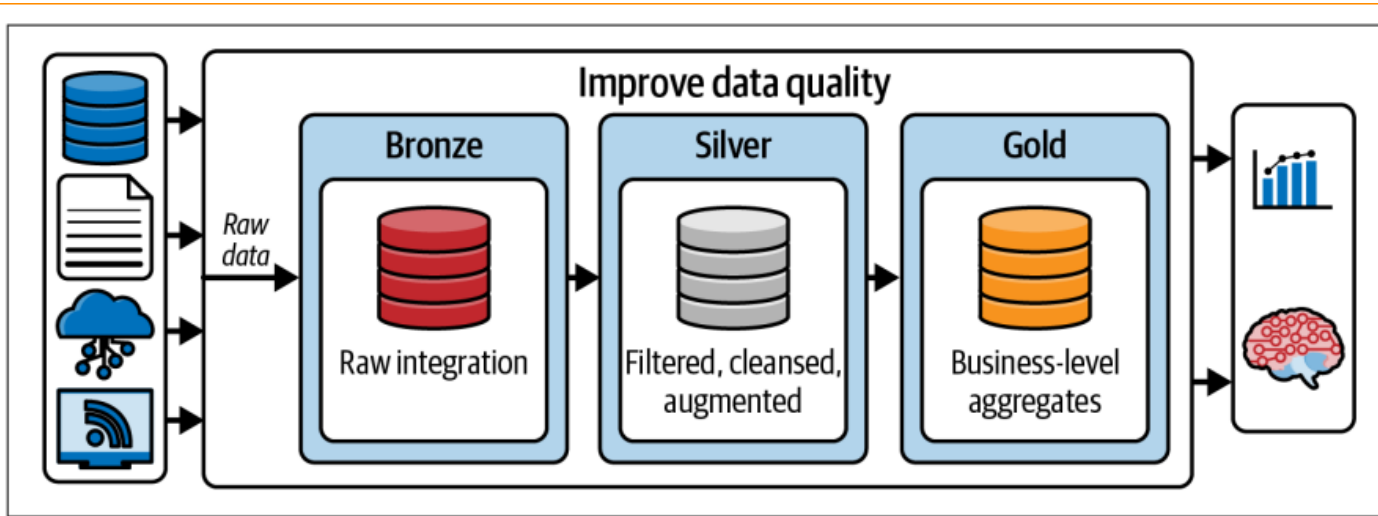


Figure 2-1. A medallion architecture promotes three distinct layers of data preparedness

- Bronze data is raw and unfiltered, often directly from integration sources. For example, your raw tables might be written directly from an API.
- Silver data is filtered, cleaned, and adjusted. You might remove unnecessary, extraneous information, apply transformations to ensure consistency, and enrich data as necessary.
- Gold data is generally stakeholder-ready and sometimes aggregated. Your consumers (analytics, engineering) should be querying Gold tables a vast majority of the time, with only occasional need for Silver tables. Generally, production assets (visualizations, reports) should not hit anything other than Gold tables.

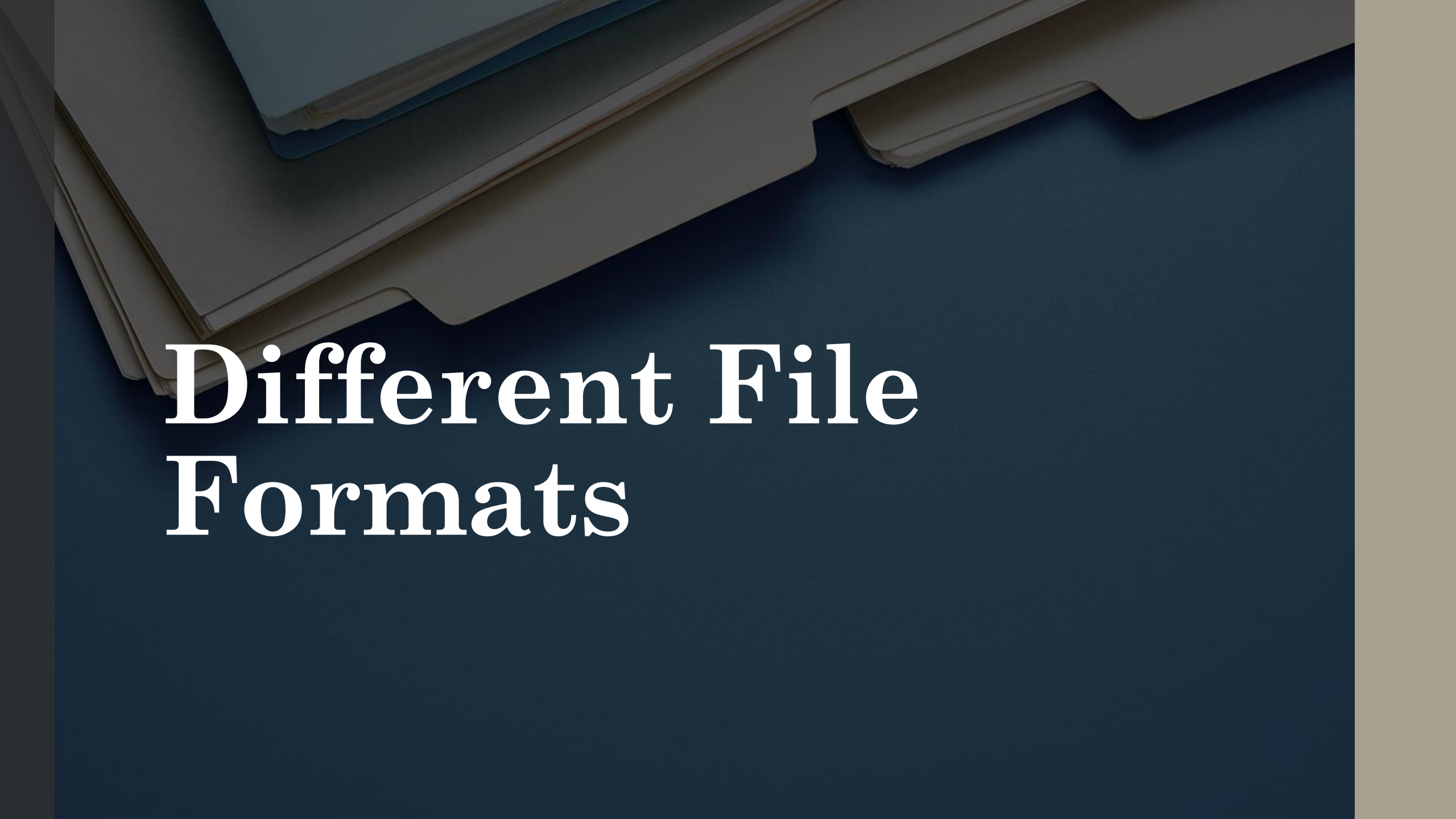
Understanding ETL

Data Pipelines for
Modern Data Architectures



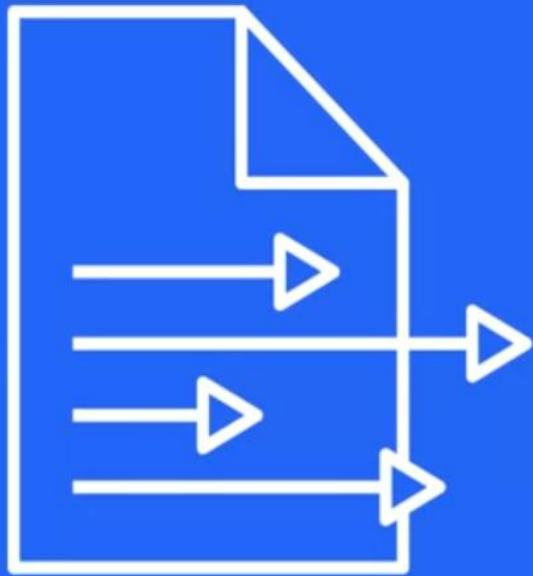
Matt Palmer

Check Canvas



Different File Formats

Delimited text files



Files used to store data as text

Each value is separated by a delimiter

Delimiter - A sequence of one or more characters for specifying the boundary between independent entities or values.

Comma, Tab, Colon, Vertical Bar, Space



Comma-separated values



Tab-separated values

Different File Formats

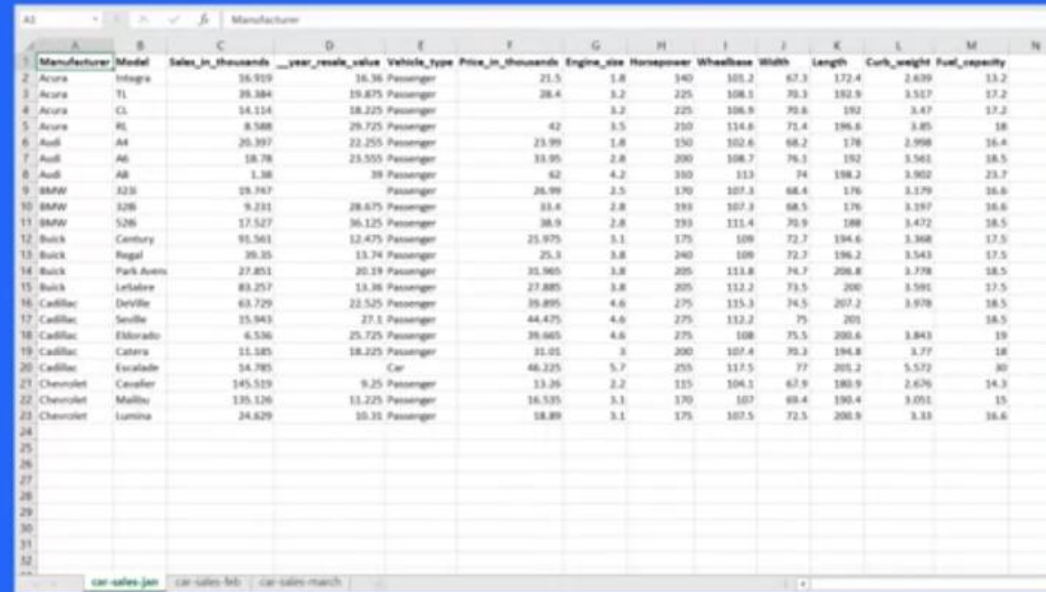
```
Manufacturer, Model, Sales_in_thousands, __year_resale_value, Vehicle_type, Price_in_thousands
Acura, Integra, 16.919, 16.36, Passenger, 21.5
Acura, TL, 39.384, 19.875, Passenger, 28.4
Acura, CL, 14.114, 18.225, Passenger, 14
Acura, RL, 8.588, 29.725, Passenger, 42
Audi, A4, 20.397, 22.255, Passenger, 23.99
Audi, A6, 18.78, 23.555, Passenger, 33.95
Audi, A8, 1.38, 39, Passenger, 62
BMW, 323i, 19.747, Passenger, 26.99
BMW, 328i, 9.231, 28.675, Passenger, 33.4
BMW, 528i, 17.527, 36.125, Passenger, 38.9
Buick, Century, 91.561, 12.475, Passenger, 21.975
```

.CSV

.TSV

Manufacturer	Model	Sales_in_thousands	__year_resale_value
Acura	Integra	16.919	16.36 Passenger 21.5
Acura	TL	39.384	19.875 Passenger 28.4
Acura	CL	14.114	18.225 Passenger 14
Acura	RL	8.588	29.725 Passenger 42
Audi	A4	20.397	22.255 Passenger 23.99
Audi	A6	18.78	23.555 Passenger 33.95
Audi	A8	1.38	39 Passenger 62
BMW	323i	19.747	Passenger 26.99
BMW	328i	9.231	28.675 Passenger 33.4
BMW	528i	17.527	36.125 Passenger 38.9
Buick	Century	91.561	12.475 Passenger 21.975

Microsoft Excel Open XML Spreadsheet, or .XLSX



The image shows a screenshot of a Microsoft Excel spreadsheet. The spreadsheet has a single sheet named 'Manufacture'. The data is organized into columns with headers: 'Manufacture', 'Model', 'Sales_in_thousands', 'year_resale_value', 'Vehicle_Type', 'Price_in_thousands', 'Engine_size', 'Horsepower', 'Wheelbase', 'Width', 'Length', 'Curb_weight', and 'Fuel_capacity'. The rows contain data for various car models from different manufacturers, including Acura, Audi, BMW, Buick, Cadillac, Chevrolet, and others. The spreadsheet is displayed in a window with a blue title bar and a standard Windows interface.

Manufacture	Model	Sales_in_thousands	year_resale_value	Vehicle_Type	Price_in_thousands	Engine_size	Horsepower	Wheelbase	Width	Length	Curb_weight	Fuel_capacity
Acura	Integra	16,519	16.36	Passenger	21.5	1.8	140	101.2	67.3	172.4	2,679	13.2
Acura	TL	39,384	19.875	Passenger	28.4	3.2	225	108.1	70.3	182.9	3,517	17.2
Acura	CL	14,114	18.225	Passenger		3.2	225	106.9	70.8	182	3,47	17.2
Acura	RL	8,588	29.725	Passenger		4.2	3.5	110	114.6	71.4	186.6	18
Audi	A4	26,397	22.255	Passenger	23.99	1.8	150	102.6	68.2	178	2,998	16.4
Audi	A6	18,78	23.555	Passenger	33.95	2.8	200	108.7	75.1	192	3,561	18.5
Audi	A8	1,38		Passenger	62	4.2	330	113	74	198.2	3,902	23.7
BMW	323i	28,747		Passenger	26.99	2.5	170	107.3	68.4	176	3,179	16.6
BMW	328	9,231	28.675	Passenger	33.4	2.8	193	107.3	68.5	176	3,197	16.6
BMW	528	17,527	36.125	Passenger	38.9	2.8	193	111.4	70.9	188	3,472	18.5
Buick	Century	91,561	12.475	Passenger	25.975	5.1	175	109	72.7	194.6	3,368	17.5
Buick	Regal	36.35	13.74	Passenger	25.3	3.8	240	109	72.7	196.2	3,543	17.5
Buick	Park Avenue	27,851	20.19	Passenger	31,965	3.8	205	113.8	74.7	208.8	3,778	18.5
Buick	LeSabre	83,257	13.36	Passenger	27,885	3.8	205	112.2	73.5	200	3,591	17.5
Cadillac	Deville	63,729	22.525	Passenger	39,895	4.6	275	115.3	74.5	207.2	3,978	18.5
Cadillac	Seville	15,943	27.1	Passenger	44,475	4.6	275	112.2	75	201		18.5
Cadillac	Eldorado	4,536	25.725	Passenger	39,665	4.6	275	108	75.5	203.6	3,843	19
Cadillac	Catera	11,185	18.225	Passenger	31.01	3	200	107.4	69.3	194.8	3,77	18
Cadillac	Escalade	14,785		Car	46,225	5.7	255	117.5	77	201.2	5,572	30
Chevrolet	Cavalier	145,519	9.25	Passenger	13.26	2.2	115	104.1	67.9	180.9	2,676	14.3
Chevrolet	Malibu	135,126	11.225	Passenger	16,535	3.1	170	107	69.4	190.4	3,051	15
Chevrolet	Lumina	24,829	10.31	Passenger	18.89	3.1	175	107.5	72.5	200.9	3,33	16.6

- Open file format, accessible to most other applications
- Can use and save all functions available in excel
- Is a secure file format as it cannot save malicious code

Row storage vs Columnar storage

ROW STORAGE

id	name	age	phone_number
1	John	35	555-1234
2	Jane	28	555-5678
3	Bob	41	555-9101

❗ Row-oriented storage stores all data for a row together. To get just names, full rows must be read, even if only names are needed.

COLUMNAR STORAGE

ID	NAME	PHONE NUMBER
1	JOHN	555-1234
2	JANE	555-5678
3	BOB	555-9101

❗ Columnar storage organizes data by column instead of row. Retrieving only names reads just that column, faster than row storage.

Aspect	Row Storage	Column Storage
How data is stored	Stores all fields of a record (row) together	Stores values of each column together
Best for	Transactional workloads (OLTP) — frequent inserts, updates, deletes	Analytical workloads (OLAP) — aggregations, queries on few columns
Example systems	MySQL, PostgreSQL, SQL Server	Apache Parquet, ORC, Amazon Redshift
Query performance	Fast for reading or writing full rows	Fast for aggregations (e.g., <code>SUM</code> , <code>AVG</code>) on selected columns
Compression efficiency	Lower (mixed data types in rows)	Higher (same data type in each column)
Use case example	Bank transactions or e-commerce order processing	Business intelligence dashboards, data warehouses

Parquet File

A **Parquet file** is a columnar storage file format optimized for analytics and big data workloads.

Employee_ID	Name	Department	Salary
101	John Doe	IT	75000
102	Jane Smith	HR	60000
103	Alice Lee	IT	80000
104	Bob Brown	Marketing	55000

Advantages of Parquet

- **Efficient Storage:** Saves disk space by compressing data.
- **Columnar Storage:** Optimized for analytical workloads (e.g., querying specific columns).
- **Interoperability:** Supported by tools like Apache Spark, Hive, and Python's Pandas.

Timestamp	Sensor_ID	Temperature	Humidity
2023-01-01 12:00:00	1	23.5	40.2
2023-01-01 12:05:00	2	25.3	45.1
2023-01-01 12:10:00	1	22.8	38.5
2023-01-01 12:15:00	2	24.7	42.3

ORC (Optimized Row Columnar) File

An **ORC (Optimized Row Columnar)** file is a high-performance file format designed for big data applications. It is particularly used in Hadoop ecosystems and is optimized for read and write operations, making it suitable for large-scale analytical workloads. Like Parquet, ORC stores data in a columnar format, but it also includes additional features like compression and indexing.

Product_ID	Product_Name	Category	Units_Sold	Price
1001	Laptop	Electronics	15	1000.99
1002	Office Chair	Furniture	50	120.49
1003	Coffee Maker	Appliances	25	75.00
1004	Smart TV	Electronics	10	800.00

Aspect	ORC	Parquet
Compression	Smaller file size, efficient	Good, but slightly larger files
Indexing	Built-in row/stripe-level indexes	No built-in indexing
Read/Write	Faster reads, slower writes	Slower reads, faster writes
Use Case	Hive-based, read-heavy workloads	General-purpose big data processing

Extensible Markup Language or .XML

```
<?xml version="1.0"?>
<car-specs>

<manufacturer>Acura</manufacturer>

<model>Integra</model>

<sales_in-thousands>16.919</sales_in-thousands>

<year_resale_value>16.36</year_resale_value>

<vehicle_type>Passenger</vehicle_type>

</car-specs>
```

Extensible Markup Language, or XML, is a markup language with set rules for encoding data.

- Readable by both humans and machines
- Self-descriptive language
- Similar to .HTML in some respects
- Does not use predefined tags like .HTML does
- Platform independent
- Programming language independent

Portable Document Format or PDF

Portable Document Format, or PDF, is a file format developed by Adobe to present documents independent of application software, hardware, and operating systems.

- Can be viewed the same way on any device
- Is frequently used in legal and financial documents
- Can also be used to fill in data for forms

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FROM THE AMERICAN PEOPLE

OMB No. 0412-0004
Expiration Date: 02/29/2012

APPLICATION FOR APPROVAL OF COMMODITY ELIGIBILITY
(FORM AID 11) Transaction No. (Assigned by USAID)

TRANSACTION IDENTIFICATION

1. USAID Letter of Commodity No.	2. Payment Terms U.S. Bank Letter of Credit No. Date	Name and Address of U.S. Bank (Advising Bank)	Other Payment Terms (if any)
3. Import License No.	Date	4. Supplier's Relationship to Authorized Source Country ☐ Corporation or Partnership Organized under Source Country Laws ☐ Individual, Officer or Partner Resident of Source Country ☐ Controlled Foreign Corporation ☐ Other	
5. Supplier's Name and Address		6. Importer's Name and Address	

7. Contract
Total Amount (Funded by USAID) \$
Date

8. Shipping Plans at Time of Application

a. Partial Shipment ☐ No ☐ Yes	b. Loading Port	c. Destination Port	d. Month(s) of Shipment
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COMMODITY IDENTIFICATION

9. Schedule B 10-Digit Code(s)	10. Commodity Description, Quantity, Size	11. Unit and Unit Price, or Total FAS/FOB Vessel Price, or FCA Price (Named Port of Loading/Airport)
(4)		
(8)		
(10)		
(12)		
(14)		
(16)		

12. Commodity Condition
☐ New and Unused ☐ Used - Not Rebuilt or Reconditioned ☐ Rebuilt ☐ Reconditioned ☐ Other (Specify below)

JavaScript Object Notation or JSON

```
{  
  "Employee": [  
    {  
      "id": "1",  
      "Manufacturer": "Audi",  
      "Model": "Integra",  
    },  
    {  
      "id": "2",  
      "Manufacturer": "Buick",  
      "Model": "LeSabre",  
    },  
    {  
      "id": "3",  
      "Manufacturer": "Cadillac",  
      "Model": "Escalade",  
    }  
  ]  
}
```

JavaScript Object Notation, or JSON, is a text-based open standard designed for transmitting structured data over the web.

- Language-independent data format
- Can be read in any programming language
- Easy to use
- Compatible with a wide range of browsers
- Considered as one of the best tools for sharing data

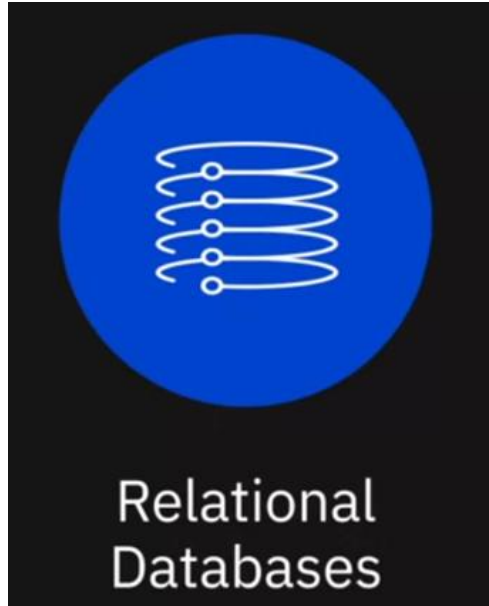
Different File Formats

File Format	Type	Storage Structure	Compression	Schema Support	Best For	Use Cases
CSV	Row-based	Plain Text	None or GZIP	No	Simple, small datasets	Log files, simple ETL tasks, data exchange
JSON	Semi-structured	Text	None or GZIP	No	Flexible schema, human-readable	API responses, configuration files, web data
Parquet	Columnar	Column-oriented	Snappy, GZIP, LZO	Yes	Large-scale analytics, high performance	Data warehouses, analytics, Spark, Presto
ORC	Columnar	Column-oriented	Zlib, Snappy, LZO	Yes	Hive and Hadoop ecosystem	Hive tables, Hadoop jobs
Avro	Row-based	Binary format	Snappy, Deflate	Yes (with schema evolution)	Streaming and batch data processing	Kafka, Hadoop, and data serialization tasks



BREAK
15MINS

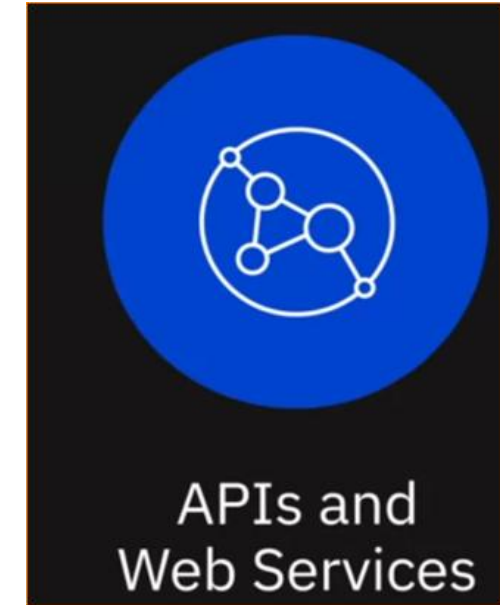
Different Data Sources?



- Organizations have internal applications to support them in managing their day to day business activities, customer transactions, human resource activities, and their workflows.
- These systems use relational databases such as SQL Server, Oracle, MySQL, and IBM DB2, to store data in a structured way.
- Data stored in databases and data warehouses can be used as a source for analysis.



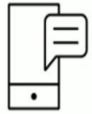
- External to the organization, there are other publicly and privately available datasets.
- E.g., government organizations releasing demographic and economic datasets on an ongoing basis.
- Then there are companies that sell specific data, for example, Point-of-Sale data or Financial data, or Weather data, which businesses can use to define strategy, predict demand, and make decisions related to distribution or marketing promotions, among other things.
- Such data sets are typically made available as flat files, spreadsheet files, or XML documents.



- APIs and Web Services typically listen for incoming requests, which can be in the form of web requests from users or network requests from applications.
- They return data in plain text, XML, HTML, JSON, or media files.
- APIs are also used for pulling data from database sources, within and external to the organization.

Different Data Sources?

Popular examples of APIs



Twitter and Facebook APIs
for customer sentiment analysis



Stock Market APIs
for trading and analysis



Data Lookup and Validation APIs
for cleaning and co-relating data

- The use of Twitter and Facebook APIs to source data from tweets and posts for performing tasks such as opinion mining or sentiment analysis—which is to summarize the amount of appreciation and criticism on a given subject, such as policies of a government, a product, a service, or customer satisfaction in general.
- Stock Market APIs used for pulling data such as share and commodity prices, earnings per share, and historical prices, for trading and analysis.
- Data Lookup and Validation APIs, which can be very useful for Data Analysts for cleaning and preparing data, as well as for co-relating data—for example, to check which city or state a postal or zip code belongs to.

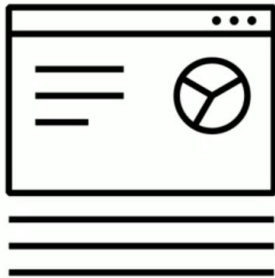
Different Data Sources?



- Extract relevant data from unstructured sources
- Also known as Screen scraping, Web harvesting, and Web data extraction
- Downloads specific data based on defined parameters
- Can extract text, contact information, images, videos, product items, and more...

Popular web scraping tools:

- BeautifulSoup
- Scrapy
- Pandas
- Selenium



Popular uses:



Providing price comparisons by collecting product details from retailer, manufacturers, and eCommerce websites



Generating sales leads through public data sources



Extracting data from posts and authors on various forums and communities



Collecting training and testing datasets for machine learning models

Feature	API	Web Scraping
Access Method	Official, structured access.	Unofficial, unstructured access.
Data Format	Structured (JSON, XML, etc.).	Unstructured (HTML, requires parsing).
Reliability	High (maintained by provider).	Low (depends on website changes).
Speed	Fast (optimized for data retrieval).	Slower (requires parsing HTML).
Legal/Ethical	Generally allowed.	May violate terms of service.
Complexity	Easy to use (if API is well-documented).	Requires more effort to implement.

API vs. Web Scraping

Different Data Sources?



Data Streams and Feeds

Aggregating streams of data flowing from instruments, IoT devices and applications, GPS data from cars, computer programs, websites, and social media posts

-  Stock and market tickers for financial trading
-  Retail transaction streams for predicting demand and supply chain management
-  Surveillance and video feeds for threat detection

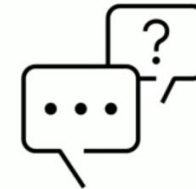
Popular technologies used to process data streams include:



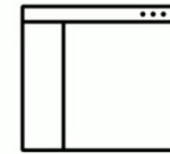
-  Social media feeds for sentiment analysis
-  Sensor data feeds for monitoring industrial or farming machinery
-  Web click feeds for monitoring web performance and improving design
-  Real-time flight events for rebooking and rescheduling

RSS (or Really Simple Syndication) feeds

Capturing updated data from online forums and news sites where data is refreshed on an ongoing basis.



Online forums



News sites

What is Data Extraction?

Data extraction is the first stage of the ETL process, where data is acquired from various source systems.

The data may be completely raw, such as sensor data from IoT devices, or perhaps it is unstructured data from scanned medical documents or company emails.

It may be streaming data coming from a social media network or near real-time stock market buy/sell transactions, or it may come from existing enterprise databases and data warehouses.

- Configuring access to data and reading it into an application
- Web scraping
- Connecting programmatically via APIs
- The data may be static or streaming online

Data Extraction Techniques



OCR



ADC sampling, CCD sampling



Mail, phone, or in-person surveys and polls



Cookies, user logs



Web scraping



APIs



Database querying



Edge computing



Biomedical devices

What is a Data Transformation?

- While data ingestion simply transfers data from point A to B, data transformation turns raw data into valuable insights through various stages of the data lifecycle
- Data transformation is the art of manipulating and enhancing data to better serve users and processes.
- Transformation involves taking some data, whether in a raw or nearly pristine state, and performing one or many operations to move it closer to the intended use.
- In an ETL pipeline, transformation occurs in not one, but many places. Data might be transformed upon ingestion and again at any number of points downstream.
- The goal of data transformation is to turn data into an asset—using analysis and science to create something of value for the business.
- This is normally done in an intermediate working environment called a “**staging area**.” Here, the data are cleaned to ensure reliability and conformed to ensure compatibility with the target system.

Data Transformation Techniques

Many other transformations may be applied, including:

- Cleaning: fixing any errors or missing values
- Filtering: selecting only what is needed
- Joining: merging disparate data sources
- Normalizing: converting data to common units
- Data Structuring: converting one data format to another, such as JSON, XML, or CSV to database tables
- Feature Engineering: creating KPIs for dashboards or machine learning
- Anonymizing and Encrypting: ensuring privacy and security
- Sorting: ordering the data to improve search performance
- Aggregating: summarizing granular data

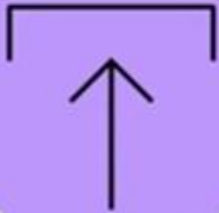
What is Data Loading?

Moving data into a new environment

Examples: A database, data warehouse, or data mart

Making the data readily available for analytics, dashboards, reports

Data Loading Strategies



Full loading used to:

- Start tracking transactions in a new data warehouse
- Load an initial history into a database



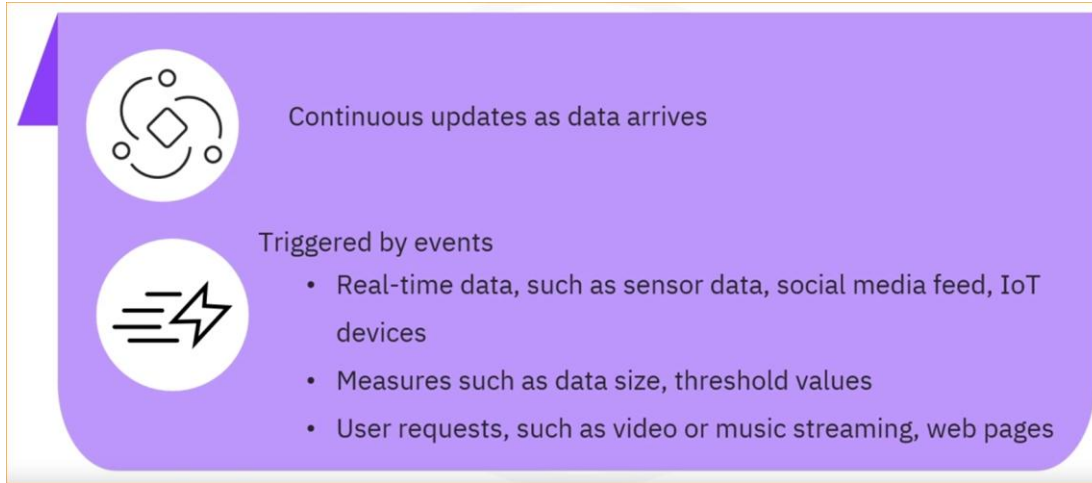
Incremental loading used to:

- Load data that has changed since previous loading
- Append to existing data
- Accumulate transaction history

Types of Incremental Loading

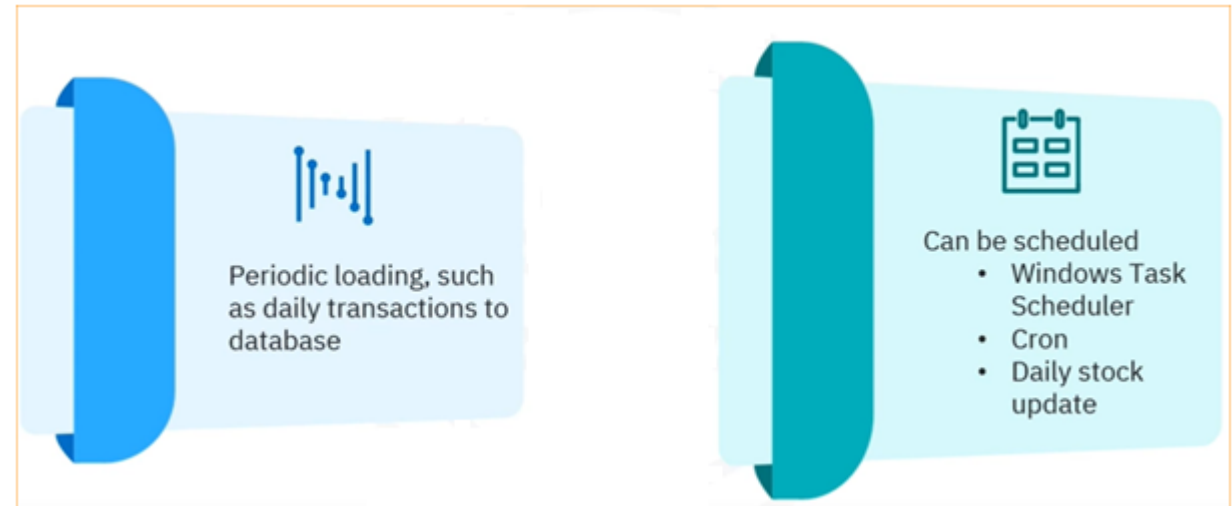
Stream Loading

Stream loading refers to the continuous loading of datasets, as they are generated.



Batch Loading

Batch loading is the act of loading data in batches rather than all at once.



Push vs. Pull Data Loading Methodology

Push

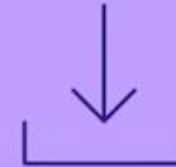
- Source pushes data to data warehouse
- Useful for real time data



Most suited for Stream Loading involving real time data

Pull

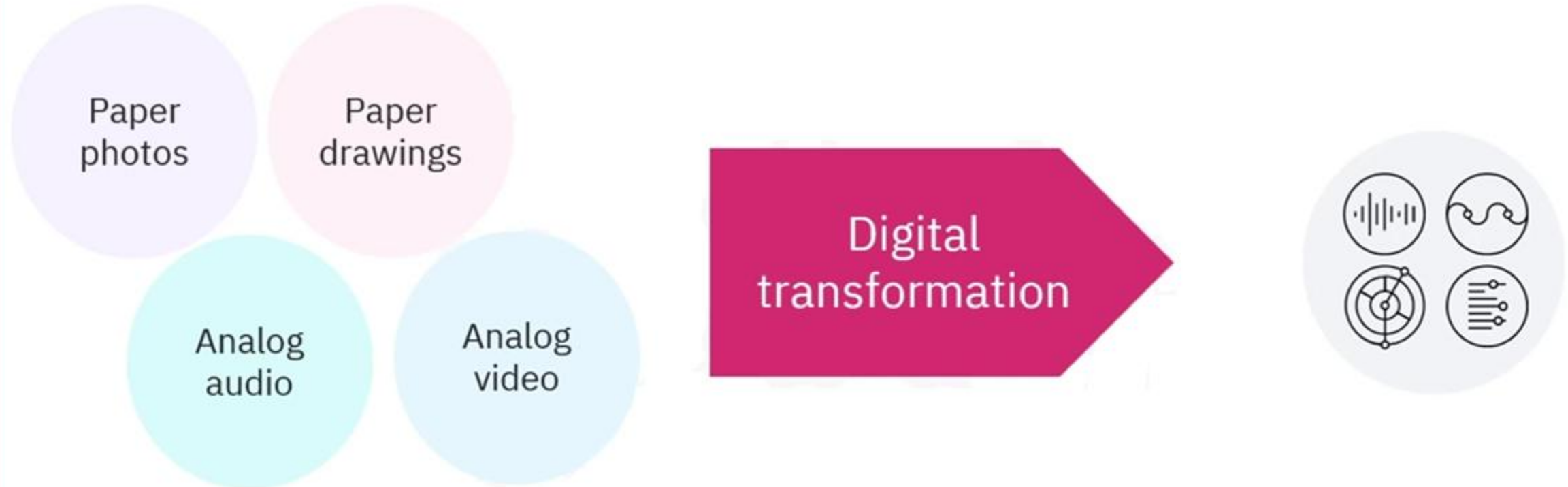
- Data warehouse pulls the data from the source
- Useful for scheduled extractions and batch loading



Mostly suited for Batch Loading

Use Cases of ETL Pipeline

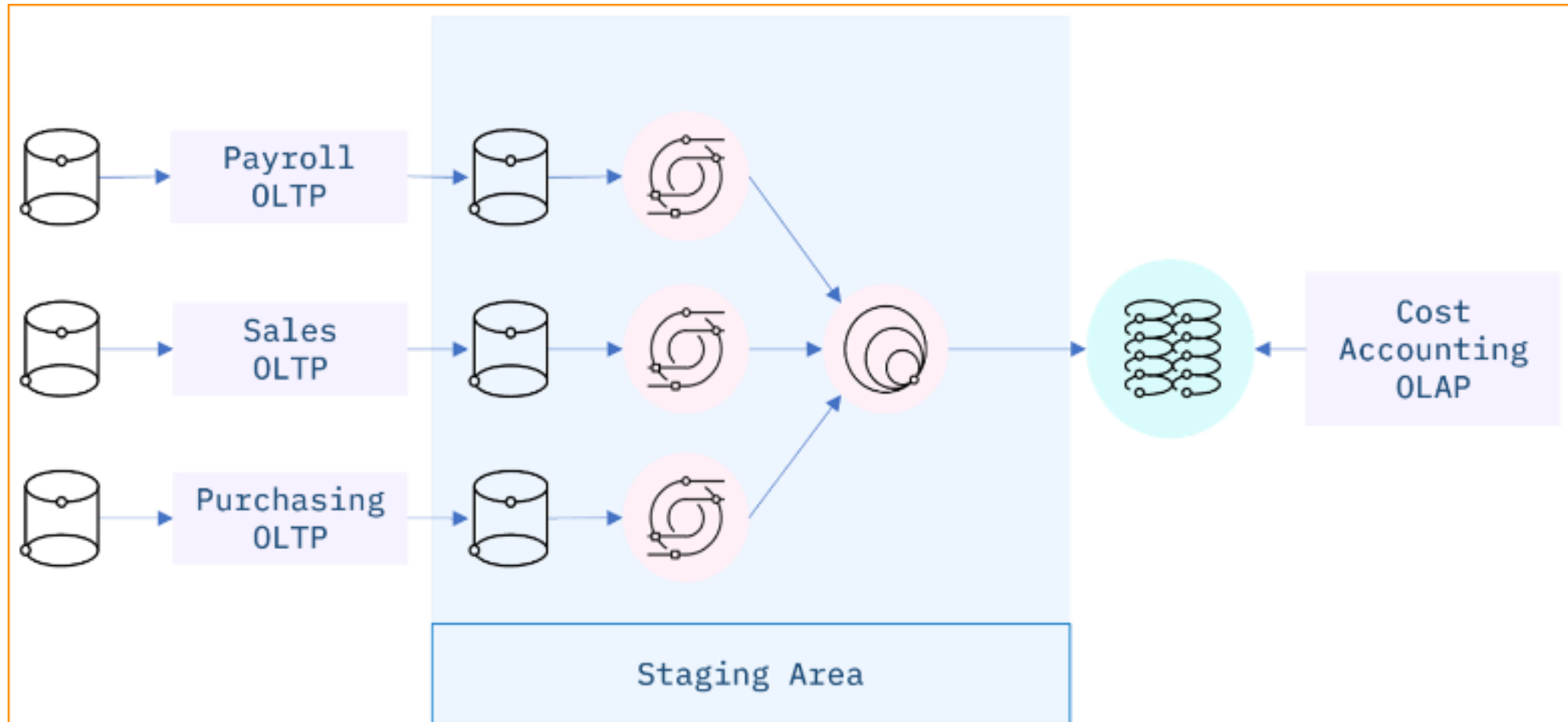
- Digitizing analog media



- Moving data from OLTP systems to OLAP systems
- Dashboards
- Machine learning

Use Cases of ETL Pipeline

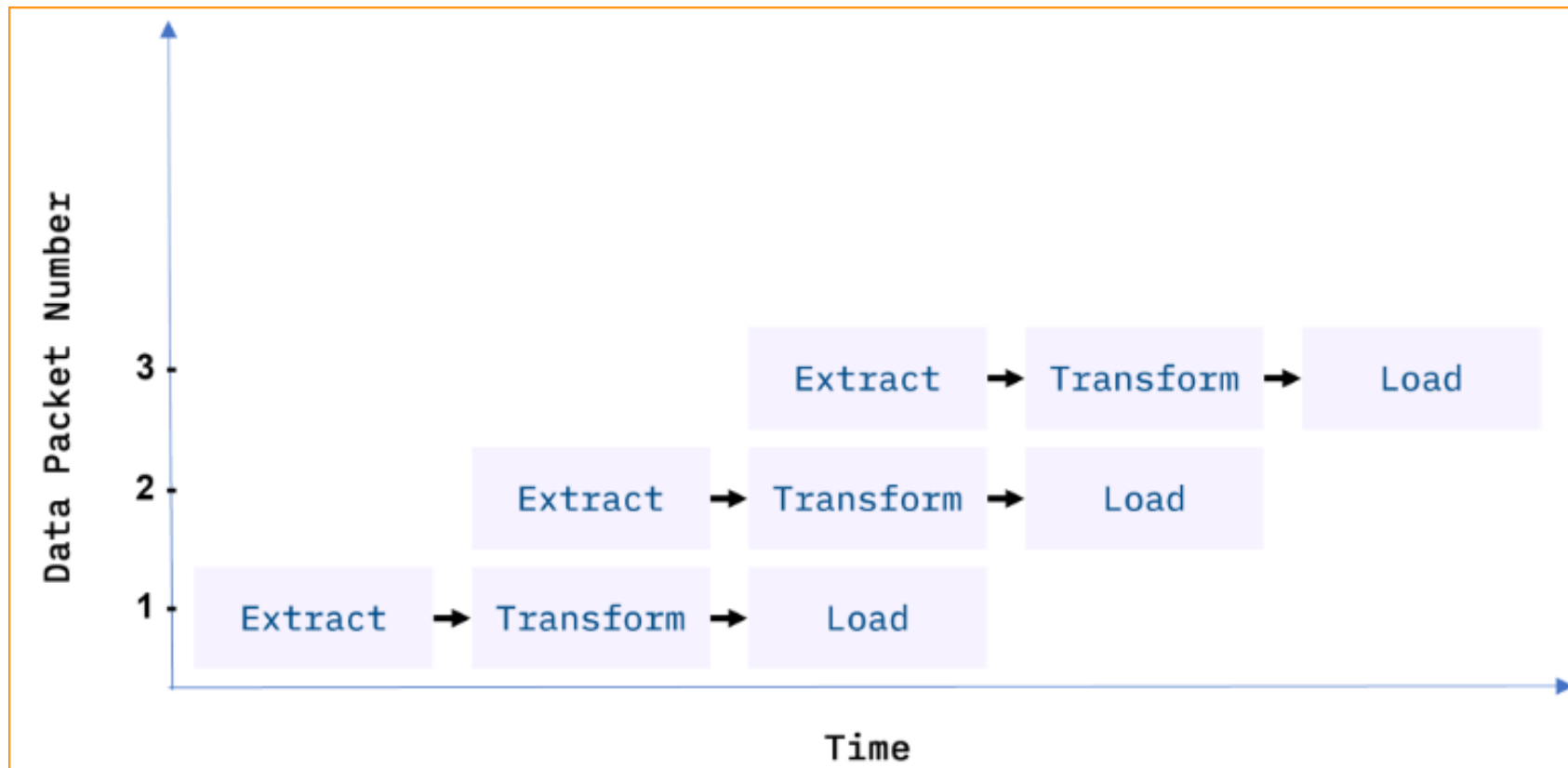
ETL pipelines are frequently used to integrate data from disparate and usually siloed systems within the enterprise. These systems can be from different vendors, locations, and divisions of the company, which can add significant operational complexity. As an example, a cost accounting OLAP system might retrieve data from distinct OLTP systems utilized by the separate payroll, sales, and purchasing departments



ETL Workflows as Data Pipelines

An ETL workflow is a well thought out process that is carefully engineered to meet technical and end-user requirements.

To boost efficiency, data is fed through a *data pipeline* in smaller packets. While one packet is being extracted, an earlier packet is being transformed, and another is being loaded. In this way, data can keep moving through the workflow without interruption. Any remaining bottlenecks within the pipeline can often be handled by parallelizing slower tasks.



Best ETL Tools

The Alteryx logo is displayed in a large, blue, lowercase sans-serif font.

Key considerations for a company's ETL tools:

- 1.The extent of data integration.
- 2.The level of customizability
- 3.The cost structure.
- 4.The level of automation provided
- 5.The level of security and compliance
- 6.The performance and reliability of the tool.



Easy to Use ETL Tool

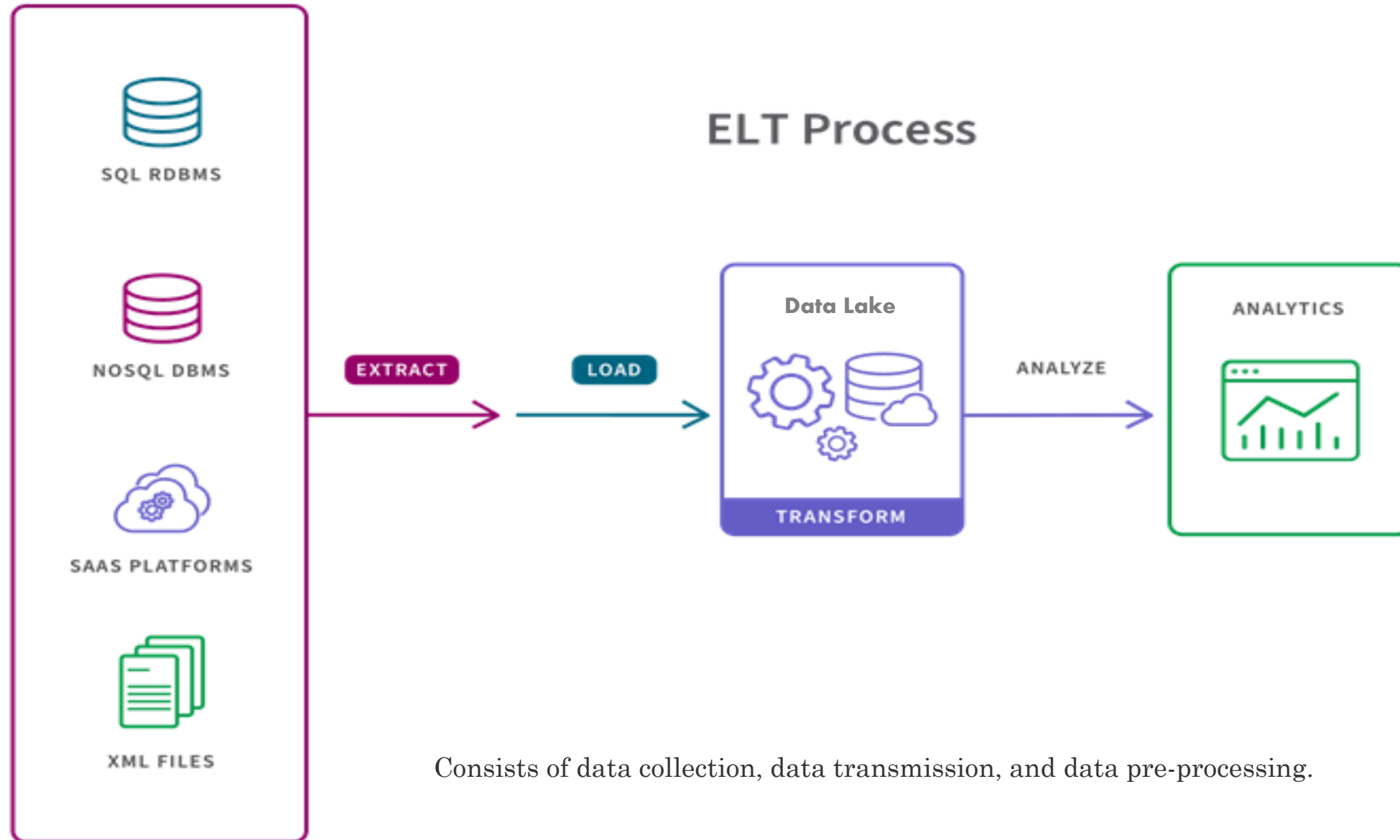
The Pandas Python library

- Versatile and popular programming tool
- Based on data frames – table-like structures
- Great for ETL, data exploration, and prototyping
- Doesn't readily scale to Big Data

Libraries with similar APIs: Vaex, Dask, and Spark help with scaling up

Consider SQL-like alternatives, such as PostgreSQL for Big Data applications

ELT Data Pipeline



Consists of data collection, data transmission, and data pre-processing.

ETL vs. ELT



When and where the transformations happen:

- Transformations for ETL happen within the data pipeline
- Transformations for ELT happen in the destination environment



Flexibility:

- ETL is rigid: pipelines are engineered to user specifications
- ELT is flexible: end users build their own transformations



Support for Big Data:

- Organizations use ETL for relational data, on-premise: scalability is difficult
- ELT solves scalability problems, handling both structured and unstructured Big Data in the cloud



Time-to-insight:

- ETL workflows take time to specify and develop
- ELT supports self-serve, interactive analytics in real-time

ETL

Structured Data

Source and destination databases are different

Complex data transformations

Data size is small

ELT

Data size is huge

Unstructured Data

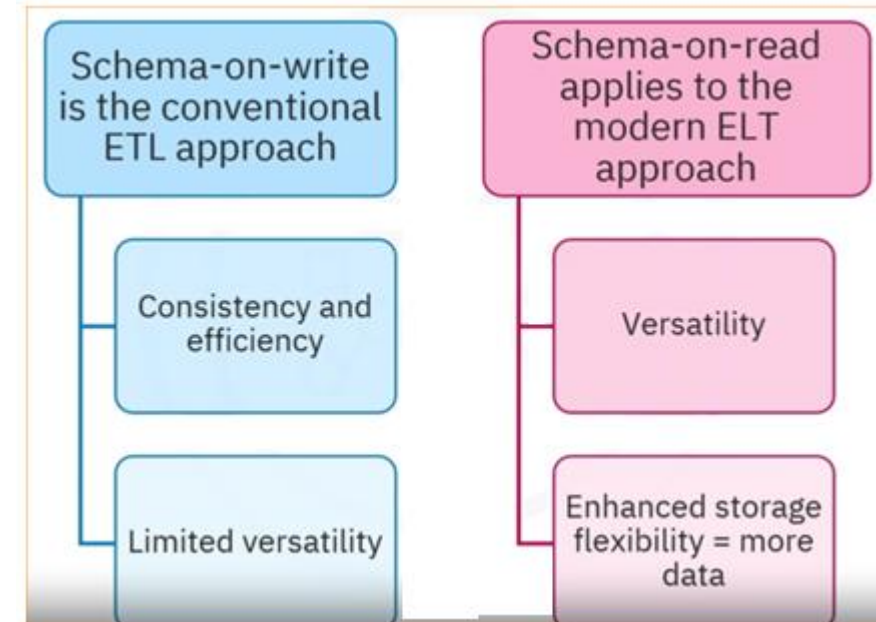
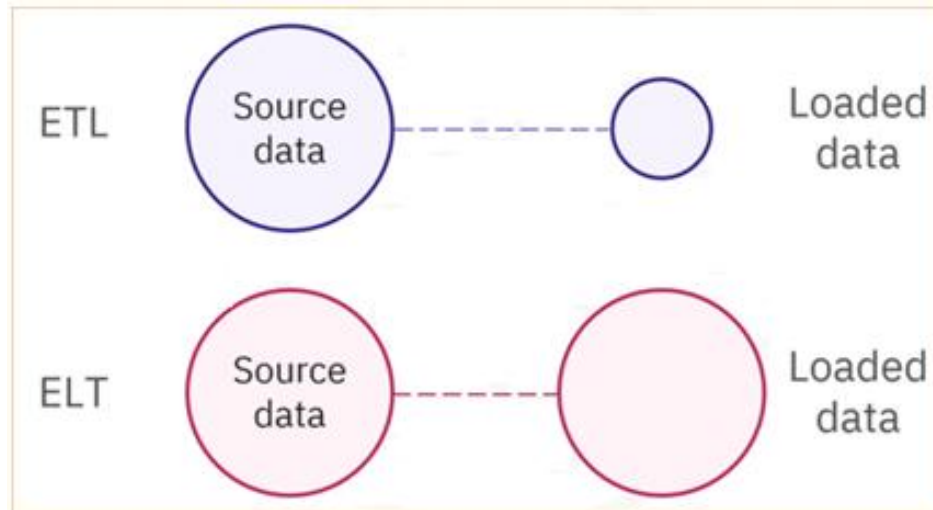
Source and destination databases are same

Simple data transformations

ELT is an Emerging Trend

ELT addresses key pain points:

- Lengthy time-to-insight
- Challenges imposed by Big Data
- Demand for access to siloed information



- ELT processes are used for cases where flexibility, speed, and scalability are important
- Cloud-based analytics platforms are ideally suited for handling Big Data and ELT processes
- ELT is an emerging trend because cloud platforms are enabling it

Trivia Time

ETL process consists of Extract > Transform > Load. Which of these three processes is also known as data wrangling?

- ☐ Extraction
- ☒ Transform
- ☐ Load
- ☐ Data wrangling is a term for another data warehouse process

Transformations for ETL happen in the data pipeline. Where do transformations happen for ELT?

- ☐ Load process
- ☐ Source environment
- ☒ Destination environment
- ☐ Extraction process

Which of the following data transformation techniques will be suitable for 'unlike' data sources?

- ☐ Cleaning
- ☐ Filtering, sorting, aggregation
- ☐ Data structuring
- ☒ Joining or merging

Trivia Time

Why is there no information loss in ELT unlike ETL where there is loss of information?

- ☒ Because the data is copied as is
- ☐ Because ELT uses edge computing
- ☐ Because ETL uses aggregation of data but ELT doesn't
- ☐ Because ELT involves lossy data compression

ETL uses the schema-on-write approach. What is the biggest disadvantage of this approach?

- ☐ Stability
- ☐ Consistency
- ☐ More data access
- ☒ Limited versatility

There are many techniques for extracting data. What does the choice of technique depend on?

- ☐ Operating system
- ☒ Kind of data source and intended use
- ☐ Type of client
- ☐ Optical or analog

Trivia Time

Which of the following pain points does ELT address?

- ☐ Request for fixed processes
- ☒ Challenges imposed by Big Data
- ☐ Cost-effectiveness
- ☐ Transformation in data pipeline

Which of the following in an ELT process best compares to the “Staging area” in the ETL process?

- ☐ Transformed data storage
- ☒ Data lake in ELT process
- ☐ Database servers
- ☐ Storage for source data in the ELT process

The ELT process has no information loss. What is the main reason for this benefit?

- ☐ Data source integration
- ☐ It separates the data pipeline from processing.
- ☐ There is a separation between moving and processing data.
- ☒ Data is acquired and directly loaded, as-is, into its destination environment.

Thank You
for
Attention

